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Vane compressor for surgical gas delivery system with gas sealed insufflation and recirculation

Abstract

A vane compressor for a surgical gas delivery system is disclosed, which includes a compressor head having an outlet port for delivering pressurized gas to a gaseous sealing manifold communicating with a gas sealed trocar and an inlet port for receiving spent gas from the gaseous sealing manifold by way of the gas sealed trocar, wherein the compressor head is coupled to and driven by a motor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The subject application is a continuation-in-part of U.S. application Ser. No. 17/155,478 filed Jan. 22, 2021, and a continuation-in-part of U.S. application Ser. No. 17/155,572 filed Jan. 22, 2021, the disclosures of which are both herein incorporated by reference in their entireties.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The subject invention is directed to endoscopic surgery, and more particularly, to a surgical gas delivery system for gas sealed insufflation and recirculation that has a vane compressor coupled to a direct current motor for use during an endoscopic or laparoscopic surgical procedure.

2. Description of Related Art

(2) Laparoscopic or “minimally invasive” surgical techniques are becoming commonplace in the performance of procedures such as cholecystectomies, appendectomies, hernia repair and nephrectomies. Benefits of such procedures include reduced trauma to the patient, reduced opportunity for infection, and decreased recovery time. Such procedures within the abdominal (peritoneal) cavity are typically performed through a device known as a trocar or cannula, which facilitates the introduction of laparoscopic instruments into the abdominal cavity of a patient.

(3) Additionally, such procedures commonly involve filling or “insufflating” the abdominal cavity with a pressurized fluid, such as carbon dioxide, to create an operating space, which is referred to as a pneumoperitoneum. The insufflation can be carried out by a surgical access device, such as a trocar, equipped to deliver insufflation fluid, or by a separate insufflation device, such as an insufflation (veress) needle. Introduction of surgical instruments into the pneumoperitoneum without a substantial loss of insufflation gas is desirable, in order to maintain the pneumoperitoneum.

(4) During typical laparoscopic procedures, a surgeon makes three to four small incisions, usually no larger than about twelve millimeters each, which are typically made with the surgical access devices themselves, often using a separate inserter or obturator placed therein. Following insertion, the obturator is removed, and the trocar allows access for instruments to be inserted into the abdominal cavity. Typical trocars provide a pathway to insufflate the cavity, so that a surgeon has an open interior space in which to work.

(5) The trocar must also provide a way to maintain the pressure within the cavity by sealing between the trocar and the surgical instrument being used, while still allowing at least a minimum amount of freedom of movement for the surgical instruments. Such instruments can include, for example, scissors, grasping instruments, and occluding instruments, cauterizing units, cameras, light sources and other surgical instruments. Sealing elements or mechanisms are typically provided on trocars to prevent the escape of insufflation gas from the abdominal cavity. These sealing mechanisms often comprise a duckbill-type valve made of a relatively pliable material, to seal around an outer surface of surgical instruments passing through the trocar.

(6) SurgiQuest, Inc., a wholly owned subsidiary of ConMed Corporation has developed unique gas sealed surgical access devices that permit ready access to an insufflated surgical cavity without the need for conventional mechanical valve seals, as described, for example, in U.S. Pat. Nos. 7,854,724 and 8,795,223. These access devices are constructed from several nested components including an inner tubular body portion and a coaxial outer tubular body portion. The inner tubular body portion defines a gas sealed central lumen for introducing conventional laparoscopic or endoscopic surgical instruments to the surgical cavity of a patient and the outer tubular body

portion defines an annular lumen surrounding the inner tubular body portion for delivering insufflation gas to the surgical cavity of the patient and for facilitating periodic sensing of abdominal pressure.

(7) SurgiQuest has also developed multimodal surgical gas delivery systems for use with the unique gas sealed access devices described above. These gas delivery systems, which are disclosed for example in U.S. Pat. Nos. 9,199,047 and 9,375,539 have a first mode of operation for providing gas sealed access to a body cavity, a second mode of operation for performing smoke evacuation from the body cavity, and a third mode of operation for providing insufflation gas to the body cavity. The first and second modes utilize a recirculating gas flow that is pressurized by a relatively large and heavy piston-type compressor that is driven by an AC motor at speeds up to 3600 rpm. Although, commonly assigned U.S. Pat. No. 10,702,306 describes a similar gas delivery system where a DC motor is coupled to a compressor head.

(8) Small diameter electric motors inherently exhibit less unbalance than larger motors due to their lower rotating inertia. This enables a small diameter electric motor to operate at substantially higher speeds than larger diameter motors without exhibiting audible noise and shorter bearing life. When used as a positive displacement compressor or pump driver, small diameter motors yield higher revolutions per given unit of time, reducing the required volumetric displacement per revolution of the compressor or pump head to produce an equivalent flow rate.

(9) For example, a prior art motor operating at 1800 rpm and coupled to a two piston pump produces a total flow rate of 45 slpm, equating to a volumetric displacement of 25 cc/rev. In contrast, a motor operating at 18,000 rpm coupled to a vane pump producing 45 slpm of flow, equates to a volumetric displacement of 2.5 cc/rev. This displacement is one-tenth that of the prior example, and can be assumed to translate into a compressor having dimensions of the cube root of one-tenth of the prior example. Or in other words, at ten times the motor speed, the compressor length, height, and width would be about 46% of the dimension of the lower speed motor.

(10) Additionally, the smaller size of the motor and head have a smaller surface area, reducing the amount of ambient air displaced through surface vibration. This will result in less audible noise that is often distracting to operating room staff. Also, the smaller mass of the reciprocating vanes compared to reciprocating pistons in the prior art pump produces less vibration and therefore less noise.

(11) Furthermore, the higher rotational speeds and vane counts produce a high frequency noise in the range of 1 to 2 kHz, which is much higher than the primary mode noise produced by a piston pump, which is typically at a frequency of 150 Hz. Equipment producing noise in the range of 100 to 400 Hz is generally perceived to be distracting and of low noise quality. Applying this concept to the example above, an 1800 rpm motor driving a two piston pump would likely produce a noise frequency of 60 Hz ($1800 \text{ rpm} \times 1 \text{ min}/60 \text{ sec} \times 2 \text{ pistons}$). Comparatively, an 18,000 rpm motor driving a six vane pump would likely produce a noise frequency of 1800 Hz ($18,000 \text{ rpm} \times 1 \text{ min}/60 \text{ sec} \times 6 \text{ vanes}$).

(12) It would be beneficial therefore to incorporate a motor driven vane compressor into a gas delivery system used during an endoscopic or laparoscopic surgical procedure, so as to reduce the size and weight of the system, and thereby improve operating room workflow and reduce the incidence of staff injury.

SUMMARY OF THE DISCLOSURE

(13) The subject invention is directed to a new and useful vane compressor assembly for a surgical gas delivery system that includes a compressor head having an outlet port for delivering pressurized gas to a gaseous sealing manifold communicating with a gas sealed trocar and an inlet port for receiving spent gas from the gaseous sealing manifold by way of the gas sealed trocar. The compressor head is coupled to and driven by a motor. The motor can be a direct current (DC) motor or an alternating current (AC) motor, but a DC motor will be relatively smaller and lighter than an AC motor, and therefore more advantageous from a manufacturing standpoint.

(14) A pump body is operatively associated with the compressor head and it defines a cylindrical bore having a central axis. A hub is mounted for rotation within the cylindrical bore of the pump body about a shaft defining an axis of rotation of the hub. The axis of rotation of the hub is offset from a central axis of the cylindrical bore, and the hub contains a plurality of circumferentially spaced apart vane slots each containing a freely sliding vane.

(15) Each end of the shaft is supported in the pump body by a bearing and a mechanical shaft seal dynamically seals a gap between the shaft and the compressor head. An inlet conduit connects the inlet port to an inlet recessed surface and an outlet conduit connects the outlet port to an outlet recessed surface, and wherein the inlet and outlet recessed surfaces are disposed about the central axis of the pump body with the inlet recess terminating at about the point where a gap between the hub and the pump body is greatest, and the outlet recess terminating at about the point where the gap between the hub and the pump body is least.

(16) The vane compressor assembly has a nominal volumetric displacement ranging from 0.1 to 10 cc/rev, and preferably the nominal volumetric displacement is about 2 cc/rev. The vane compressor assembly has a nominal speed ranging from 5,000 to 100,000 rpm, and preferably the nominal speed is about 20,000 rpms. The hub of the compressor head has between three and twelve vanes each freely sliding within a corresponding slot, and preferably there are six vanes each freely sliding within a corresponding slots.

(17) The subject invention is also directed to a surgical gas delivery system for gas sealed insufflation and recirculation, which includes a gaseous sealing manifold for communicating with a gas sealed access port, an insufflation manifold for communicating with the gas sealed access port and with a valve sealed access port, a positive displacement rotary vane compressor for recirculating gas through the gas sealed access port by way of the gaseous sealing manifold, and a direct current motor coupled to the vane compressor.

(18) These and other features of the vane compressor of the subject invention will become more readily apparent to those having ordinary skill in the art to which the subject invention appertains from the detailed description of the preferred embodiments taken in conjunction with the following brief description of the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that those skilled in the art will readily understand how to make and use the vane compressor of the subject invention without undue experimentation, preferred embodiments thereof will be described in detail herein below with reference to the figures wherein:

(2) FIG. 1 is a schematic diagram of the multi-modal gas delivery system of the subject invention, which includes a gaseous sealing manifold for communicating with a gas sealed access port and an insufflation manifold for communicating with the gas sealed access port and with a valve sealed access port, wherein the gas delivery system includes a vane compressor assembly;

(3) FIG. 2 is a perspective view of the vane compressor assembly employed in the gas delivery system of FIG. 1, which includes a compressor head coupled to a direct current motor;

(4) FIG. 3 is a front elevational view of the vane compressor assembly with the end cap removed to show the hub housed within the cylindrical bore of the pump body;

(5) FIG. 4 is a perspective view of the rotating hub assembly shown with six circumferentially spaced apart vanes; and

(6) FIG. 5 is a cross-sectional view of the compressor head showing details of the internal porting.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(7) Referring now to the drawings wherein like reference numerals identify similar structural elements and features of the subject invention, there is illustrated in FIG. 1 a new and useful multi-

modal surgical gas delivery system **10** that is adapted and configured for gas sealed insufflation, recirculation and smoke evacuation during an endoscopic or laparoscopic surgical procedure. The multi-modal surgical gas delivery system **10** of the subject invention includes a gaseous sealing manifold **110** for communicating with a gas sealed access port **20** and an insufflation manifold **210** for communicating with the gas sealed access port **20** and with a valve sealed access port **30**.

(8) The gas sealed access port **20** is of the type disclosed in commonly assigned U.S. Pat. No. 8,795,223, which is incorporated herein by reference. The gas sealed access port **20** is adapted and configured to provide gas sealed instrument access to a body cavity, while maintaining a stable pressure within the body cavity (e.g., a stable pneumoperitoneum in the peritoneal or abdominal cavity). In contrast, the valve sealed access port **30** is a conventional or standard trocar, for providing access to a body cavity through a mechanical valve seal, such as, for example, a duckbill seal, septum seal or the like. Depending upon the requirements of a particular surgical procedure, the multi-modal gas delivery system **10** can be utilized with either the gas sealed access port **20**, the valve sealed access port **30** or with both access ports **20**, **30** at the same time.

(9) The gas delivery system **10** further includes a compressor assembly or positive displacement pressure pump **40** for recirculating surgical gas through the gas sealed access port **20** by way of the gaseous sealing manifold **110**. The compressor **40** is preferably driven by a brushless DC (direct-current) motor, which can be advantageously controlled to adjust gas pressure and flow rates within the gas delivery system **10**. Alternatively, the compressor **40** can be driven by an AC motor, but a DC motor will be relatively smaller and lighter, and therefore more advantageous from a manufacturing standpoint. A preferred embodiment of a positive displacement rotary vane compressor coupled to a direct current motor is described in greater detail below with reference to FIGS. 2 through 5.

(10) An intercooler and/or condenser **50** is operatively associated with the compressor **40** for cooling or otherwise conditioning gas recirculating through the gaseous sealing manifold **110**. A UVC irradiator **52** is operatively associated with the intercooler or condenser **50** for sterilizing gas recirculating through the internal flow passages **54** formed therein by way of the compressor **40**. In addition, the UVC irradiator is intended to sterilize the interior surfaces of the gas conduits or flow passages through which the gas flows within the intercooler/condenser **50**.

(11) The UVC irradiator **52** preferably includes at least one LED light source or a florescent light source that is adapted and configured to generate UVC radiation at a wavelength of about between 240-350 nm, and preferably about 265 nm. This ultraviolet light at such a wavelength can sterilize viral, bacterial and microbial bodies within the gas conduits of the system, and can reduce coronavirus including SARS-COV-2.

(12) Preferably, compressor assembly **40**, intercooler/condenser **50**, gaseous sealing manifold **110** and insufflation manifold **210** are all enclosed within a common housing, which includes a graphical user interface and control electronics, as disclosed for example in commonly assigned U.S. Pat. No. 9,199,047, which is incorporated herein by reference.

(13) The gas delivery system **10** further includes a surgical gas source **60** that communicates with the gaseous sealing manifold **110** and the insufflation manifold **210**. The gas source **60** can be a local pressure vessel or a remote supply tank associated with a hospital or healthcare facility. Preferably, gas from the surgical gas source **60** flows through a high pressure regulator **65** and a gas heater **70** before it is delivered to the gaseous sealing manifold **110** and the insufflation manifold **210**. Preferably, the high pressure regulator **65** and the gas heater **70** are also enclosed with the compressor **40**, intercooler **50**, gaseous sealing manifold **110** and insufflation manifold **210** in the common housing.

(14) The gas delivery system **10** further includes a first outlet line valve (OLV1) **212** that is operatively associated with the insufflation manifold **210** for controlling a flow of insufflation gas to the valve sealed access port **30** and a second outlet line valve (OLV2) **214** that is operatively associated with the insufflation manifold **210** for controlling a flow of insufflation gas to the gas

sealed access port **20**.

(15) In accordance with a preferred embodiment of the subject invention, the first and second outlet line valves **212**, **214** of insufflation manifold **210** are proportional valves that are configured to dynamically alter or otherwise control the outflow of insufflation gas to the access ports **20**, **30** to match volume fluctuations that may arise in a patient's body cavity as they occur. The first and second proportional outlet line valves **212**, **214** provide the gas delivery system **10** with fine control of insufflation gas flow rate to achieve stable flow rates at lower pressure, reduce pressure oscillation and eliminate pneumatic hammer.

(16) Because the first and second proportional outlet line valves **212**, **214** are proximal to the patient where flow friction losses are relatively low, the gas delivery system **10** is able to measure peritoneal pressures accurately. Moreover, the use of proportional outlet line valves for this purpose is uniquely possible here, because there is constant gas recirculation throughout the gas delivery system **10**, either by way of closed loop smoke evacuation or by way of the gas sealed access port **20**.

(17) Proportional valves allow for infinitely variable gas flow adjustment between a minimum flow state and a maximum flow state. Given that some volume changes in a patient's body cavity, such as breathing, are expected and consistent, by employing proportional outlet line valves, the insufflation manifold **210** is able to dynamically alter the gas flow to the body cavity to inverse the expected volume changes, resulting in a neutral effect on the pressure inside the cavity.

(18) An additional benefit of using proportional valves for controlling the outflow of insufflation gas from manifold **210** is a reduction in response time, as compared to that of a solenoid valve. A solenoid valve operates by applying energy to coils, which produces an electromagnetic force that moves a piston. However, the energizing of the coils takes some amount of time, introducing a delay between a commanded action and the physical movement of the piston. In contrast, proportional valves, as employed in the gas delivery system **10** of the subject invention, do not have an energization delay in general, and so they have an improved response time as compared to solenoid valves.

(19) The insufflation manifold **210** further includes a first patient pressure sensor (PWS1) **222** downstream from the first outlet line valve **212** and a second patient pressure sensor (PWS1) **224** downstream from the second outlet line valve **214**. These two patient pressure sensors are used to measure abdominal pressure to control outlet line valves **212**, **214**, respectively. Two other pressure sensors are located upstream from the outlet line valves **212**, **214**, and are labeled as DPS1 and DPS2. These two pressure sensors are situated within a venturi to measure a pressure differential that is used to infer a total gas flow rate from the insufflation manifold **210** to the patient's body cavity.

(20) A primary proportional valve (PRV) **216** is also operatively associated with insufflation manifold **210** and it is located upstream from the first and second outlet line valves **212**, **214** to control the flow of insufflation gas to the first and second outlet line valves **212**, **214**. Proportional valve **216** functions to maintain an intermediate pressure within the insufflation manifold **210** (as the central node in the LPU) at a constant pressure between 1 and 80 mmHg, dependent on the system operating mode. The opening of PRV **216** can be indirectly initiated by any of the following actions: patient respiration, gas leakage downstream of PRV **216**, or the opening of the safety valve LSV **227** or ventilation valve VEV **228**, i.e. any event that causes an intermediate pressure to drop. In the system. LSV **227** and VEV **228** are described in more detail below.

(21) The gaseous sealing manifold **110** also includes a high pressure gas fill valve (GFV) **112** that is operatively associated with an outlet side of the compressor **40**. GFV **112** is adapted and configured to control gas delivered into the gaseous sealing manifold **110** from the source of surgical gas **60**. Preferably, the gas fill valve **112** is a proportional valve that is able to dynamically control surgical gas delivered into the gaseous sealing manifold **110**.

(22) The gaseous sealing manifold **110** also includes a smoke evacuation valve (SEV) **114** that is

operatively associated with an outlet side of the compressor **40** for dynamically controlling gas flow between the gaseous sealing manifold **110** and the insufflation manifold **210** under certain operating conditions, such as, for example, when the gas delivery device **10** is operating in a smoke evacuation mode. Preferably, the smoke evacuation valve **114** is a proportional valve.

(23) A bypass valve (SPV) **116** is positioned between an outlet side of the compressor **40** and an inlet side of the compressor **40** for controlling gas flow within the gaseous sealing manifold **110** under certain operating conditions. Preferably, the bypass valve **116** is a proportional valve, which is variably opened to establish and control the gaseous seal generated within gas sealed access port **20**. Moreover, bypass valve **116** controls gas flow rate to the gaseous seal using feedback from pressure sensors **122**, **124**, described in further detail below.

(24) The gaseous sealing manifold **110** also includes an air ventilation valve (AVV) **118**, which is operatively associated with an inlet side of the compressor **40** for controlling the entrainment of atmospheric air into the system **10** under certain operating conditions. For example, AVV **118** will permit the introduction of atmospheric air into the gaseous sealing circuit to increase the air mass (i.e., the standard volume) within the circuit. The thermodynamics of clinical use conditions can cause a loss of standard volume within the gas circuit. The ventilation valve **118** permits the gas delivery system **10** to make up for this lost volume, in order to ensure that pump pressure and flow rates are sufficient to maintain the gaseous seal within the gas sealed access port **20**. The ventilation valve **118** can also be opened to reduce the vacuum side pressure in the gas seal circuit.

(25) An overpressure relief valve (ORV) **120** is operatively associated with an outlet side of the compressor **40** for controlling a release of gas from the system **10** to atmosphere under certain operating conditions. Preferably, the overpressure relief valve **120** is a proportional valve that is opened to reduce the positively pressurized side of the gas seal circuit, especially in the event of an emergency, such as a loss of power to the gas delivery system **10**. The normally open configuration of relief valve **120** reduces the risk of over-pressurization of the patient cavity upon loss of power to that valve.

(26) A first pressure sensor (RLS) **122** is operatively associated with an inlet side of the compressor **40** and a second pressure sensor (PLS) **124** is operatively associated with an outlet side of the compressor **40**. These pressure sensors **122**, **124** are situated to have unobstructed and minimally restricted commutation with the patient's abdominal cavity in order to continuously and accurately measure cavity pressure. The signals from these two pressure sensors **122**, **124** are employed by a controller of the gas delivery system **10** to modulate the opening of the two outlet line valves **212** and **214**, to control the patient cavity pressure.

(27) In addition, the gaseous sealing manifold **110** includes a gas quality sensor **126** that is operatively associated with an outlet side of the compressor **40**. The gas quality sensor monitors the level of oxygen in the recirculation circuit, which corresponds to a concentration of CO₂ in the body cavity of a patient, as disclosed in U.S. Pat. No. 9,199,047.

(28) A first blocking valve (BV1) **132** is operatively associated with an outlet flow path of the gaseous sealing manifold **110** and a second blocking valve (BV2) **134** is operatively associated with an inlet flow path to the gaseous sealing manifold **110**. The blocking valves **132**, **134** are employed during a self-test prior to a surgical procedure, as disclosed in U.S. Pat. No. 9,199,047. It is envisioned that the first and second blocking valves **132**, **134** could be mechanically actuated or pneumatically actuated.

(29) A first filter element **142** is positioned downstream from the first blocking valve **132** for filtering pressurized gas flowing from the compressor **40** to the gas sealed access port **20**, and a second filter element **144** is positioned upstream from the second first blocking valve **134** for filtering gas returning to the compressor **40** from the gas sealed access port **20**. Preferably, the filter elements **142**, **144** are housed within a common filter cartridge, as disclosed for example in U.S. Pat. No. 9,199,047.

(30) The first and second blocking valves **132**, **134** communicate with a blocking valve pilot (BVP)

226 that is included within with the insufflation manifold **210**. Preferably, the blocking valve pilot **226** is a solenoid valve. It is envisioned that BVP **226** could be fed from the compressor outlet as shown or from a gas source such of surgical gas or air. The insufflation manifold **110** further includes a pressure sensor (PMS) **225** located downstream from the primary proportional valve **216** and upstream from the outlet line valves **212**, **214**. The two outlet line valves are opened to introduce insufflation gas to the patient's body cavity by way of the access ports **23**, **30**. This introduction of gas has the effect of increasing pressure within the body cavity. Additionally, the outlet line valves **212**, **214** can be opened in conjunction with air ventilation valve **228** to release gas from the body cavity, having the effect of desufflation and reduction of cavity pressure.

(31) The insufflation manifold **210** further includes a low pressure safety valve (LSV) **227** downstream from the primary proportional valve **216** and upstream from the first and second outlet line valves **212**, **214** for controlling a release of gas from the system **10** to atmosphere under certain operating conditions. LSV **227** is a purely mechanical valve that functions to limit the maximum intermediate pressure within the manifold **210** or LPU (Low Pressure Unit) in the event of a power interruption, a pressure controller malfunction or if a valve located upstream from the LSV sticks in an open position.

(32) In addition, a ventilation exhaust valve (VEV) **228** is positioned downstream from the primary proportional valve **216** and upstream from the outlet line valves **212**, **214** for controlling a release of gas from the system **10** to atmosphere under certain operating conditions. The ventilation exhaust valve **228** is a preferably a proportional valve that is opened to de-sufflate or otherwise reduce patient cavity pressure. Additionally, VEV **228** can be opened to reduce intermediate pressure within the LPU.

(33) A filter element **242** is positioned downstream from the first outlet line valve **212** for filtering insufflation gas flowing from the insufflation manifold **210** to the valve sealed access port **30**. Another filter element **244** is positioned downstream from the second outlet line valve **224** for filtering insulation gas flowing from the insufflation manifold **210** to the gas sealed access port **20**. Preferably, filter element **244** is housed with filter elements **142** and **144** in a common filter cartridge, while filter element **242** is separately located.

(34) Referring now to FIG. 2, there is illustrated the positive displacement rotary vane compressor assembly **40** of the subject invention. The compressor assembly **40** includes a direct current (DC) motor **300** that is directly coupled to a compressor head **400**. Alternatively, an alternating current (AC) motor could be coupled to the compressor head **400**, but a DC motor will be relatively smaller and lighter than an AC motor, and therefore more advantageous from a manufacturing standpoint. In either case, the motor **300** is driven by an electrical controller. During operation, gas is induced into compressor head **400** through an inlet port **412** and discharged through an outlet port **414**. More particularly, the outlet port **414** delivers pressurized gas to the gaseous sealing manifold **110** which communicates with the gas sealed trocar **20** and the inlet port **412** receives spent gas from the gaseous sealing manifold **110** by way of the gas sealed trocar **20**, as best seen in FIG. 1.

(35) In accordance with the subject invention, the compressor assembly **40** has a nominal volumetric displacement ranging from 0.1 to 10 cc/rev and preferably the nominal volumetric displacement of the compressor assembly **40** is about 2 cc/rev. The compressor assembly **40** rotates at a nominal radial speed ranging from 5,000 to 100,000 rpm and preferably the nominal radial speed of the compressor assembly **40** is about 20,000 rpm.

(36) Referring now to FIG. 3, there is illustrated an end view of the compressor assembly **40** with the end cap **418** of the pump body **420** removed to show the hub **422** of the compressor assembly **40** situated therein. More particularly, the hub **422** rotates within a cylindrical bore **424** of the pump body **420**. The axis of rotation of the hub **422** is offset from the cylindrical bore **424** of the pump body **420**.

(37) Referring to FIG. 4, the hub **422** contains a plurality of equally spaced slots **426** in which

respective vanes **428** are free to slide. The nominal design of the compressor assembly **40** of the subject invention employs six vanes **428** sliding within six respective slots **426** of a common hub **422**, although other embodiments of the compressor assembly **40** may include between 3 and 12 vanes inclusively.

(38) The vane slots **426** need not radially intersect the axis of the hub **422**, instead they may be offset from the axis of the hub **422**. Centripetal force due to the rotation of the hub **422** keeps each vane in contact with the interior wall of the cylindrical bore **424** of pump body **420** forming a dynamic seal therebetween. As the hub **422** rotates and the vanes **428** maintain contact with the interior wall of the bore **424**, the sealed volume between each adjacent pair of vanes **428** changes. During one half of the rotation, volume is increasing and drawing gas in through the inlet port **412**, and during the other half of the rotation, the volume is decreasing and compressing gas and moving it out of the outlet port **414**.

(39) With continuing reference to FIG. **4**, the hub **422** is connected to and rotates with an elongated shaft **430**. Each end of the shaft **430** is supported by a bearing **432**. A mechanical shaft seal **434** dynamically seals the gap between the shaft **430** and the compressor head **400**.

(40) Referring now to FIG. **5**, there is shown the internal porting features of the pump body **420**. More particularly, an inlet conduit **436** connects the inlet port **412** to an inlet recessed surface **438**. Similarly, an outlet conduit **440** connects the outlet port **414** to an outlet recessed surface **442**. The recessed surfaces **438**, **442** are disposed about the central axis of the cylindrical bore **424** of pump housing **420** with the inlet recess **438** terminating at about the point where the gap between the hub **422** and the pump body **420** is greatest, and the outlet recess **442** terminating at about the point where the gap between the hub **422** and the pump body **420** is least.

(41) While the vane compressor assembly and gas delivery system of the subject disclosure has been shown and described with reference to preferred embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the scope of the subject disclosure.

Claims

1. A surgical gas delivery system comprising: a) a manifold for communicating with a gas sealed access port; b) a proportional fill valve for dynamically controlling delivery of a gas into the manifold from a gas source; c) a positive displacement rotary vane compressor assembly for recirculating the gas through the gas sealed access port by way of the manifold, the positive displacement rotary vane compressor assembly including a compressor head having an inlet port on a first side surface of the compressor head for receiving a spent gas flow from the gas sealed access port through the manifold and an outlet port on a second side surface of the compressor head for discharging a pressurized gas flow to the gas sealed access port through the manifold; d) a pump body extending from a front surface of the compressor head; e) a brushless direct current motor coupled to a rear surface of the compressor head for driving the positive displacement rotary vane compressor assembly; and f) an intercooler and/or a condenser interposed between the manifold and the positive displacement rotary vane compressor assembly for conditioning the spent gas flow received by the positive displacement rotary vane compressor assembly from the manifold and for conditioning the pressurized gas flow discharged from the positive displacement rotary vane compressor assembly to the manifold, wherein the spent gas flow travels directly from an outlet of the manifold to a first inlet of the intercooler and/or the condenser and then from a first outlet of the intercooler and/or the condenser to the inlet port of the compressor head, and the pressurized gas flow travels directly from the outlet port of the compressor head to a second inlet of the intercooler and/or the condenser and then from a second outlet of the intercooler and/or the condenser to an inlet of the manifold.

2. The surgical gas delivery system as recited in claim 1, wherein the pump body defines a

cylindrical bore, and a hub is mounted for rotation within the cylindrical bore of the pump body about a shaft that defines an axis of rotation of the hub, and wherein the axis of rotation of the hub is offset from a central axis of the cylindrical bore, and wherein the hub contains a plurality of circumferentially spaced apart vane slots each containing a freely sliding vane.

3. The surgical gas delivery system as recited in claim 2, wherein each end of the shaft is supported within the pump body by a bearing, and a mechanical shaft seal dynamically seals a gap between the shaft and the compressor head.

4. The surgical gas delivery system as recited in claim 1, wherein the positive displacement rotary vane compressor assembly has a nominal volumetric displacement ranging from 0.1 to 10 cc/rev.

5. The surgical gas delivery system as recited in claim 1, wherein the positive displacement rotary vane compressor assembly has a nominal volumetric displacement of about 2 cc/rev.

6. The surgical gas delivery system as recited in claim 1, wherein the positive displacement rotary vane compressor assembly has a nominal speed ranging from 5,000 to 100,000 rpm.

7. The surgical gas delivery system as recited in claim 1, wherein the positive displacement rotary vane compressor assembly has a nominal speed of about 20,000 rpms.
